

Lecture 6

More on k -means, toward density estimation

COURSE MATH 637: Mathematical Techniques in Data Science

WEBSITE <https://jacobcalvert.com/teaching/math637fall2026/>

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OFFICE HOURS MW 2:45pm–3:45pm, Ewing Hall 509

Announcements & Reminders

- Homework 1 is due before class on Sep 14

Plan for today

- | | |
|-------------------------------|--------|
| • Recap of k -means | 7 min |
| • Example: Color quantization | 8 min |
| • Issues with k -means | 5 min |
| • Probability review | 15 min |
| • Mixtures of Gaussians | 15 min |

Recap: k -means objective

Problem (k -means)

Given examples $x_1, \dots, x_n \in \mathbb{R}^p$ and a number of clusters k , solve

$$\arg \min_{c, \mu_1, \dots, \mu_k} J(c, \mu) = \sum_{i=1}^n \|x_i - \mu_{c(i)}\|^2,$$

over all assignments $c : \{1, \dots, n\} \rightarrow \{1, \dots, k\}$ and all $\mu_1, \dots, \mu_k \in \mathbb{R}^p$.

Recap: Lloyd's algorithm

Definition (Lloyd's algorithm)

Initialize centers $\mu_1, \dots, \mu_k$. Until the assignment stops changing, repeat:

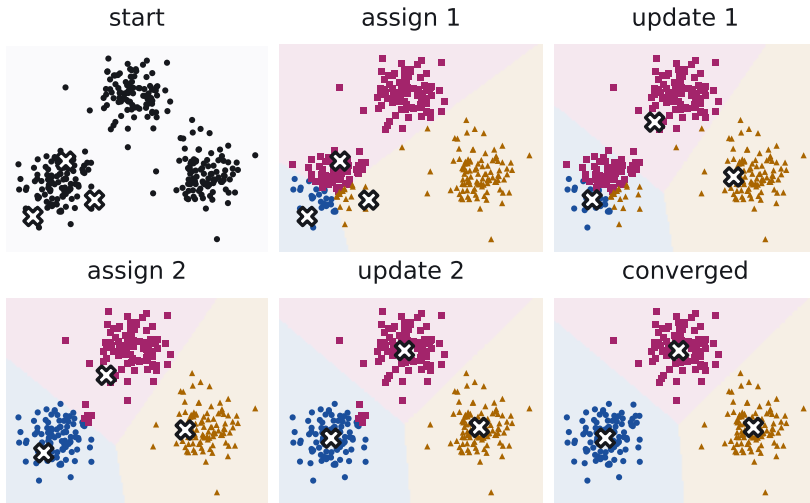
1. **Assignment.** Assign each x_i to the nearest center:

$$c(i) = \arg \min_j \|x_i - \mu_j\|^2.$$

2. **Update.** Update each μ_j to be the centroid of points assigned to it:

$$\mu_j = \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

Recap: Lloyd's algorithm



Recap: k -means++ initialization

Definition (k -means++)

1. Choose the first center μ_1 uniformly at random from $x_1, \dots, x_n$
2. Given $\mu_1, \dots, \mu_j$, set μ_{j+1} to be x_i with probability proportional to $D(x_i)^2$, where

$$D(x) = \min_{1 \leq l \leq j} \|x - \mu_l\|.$$

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Example: Color quantization



A still from *The Life and Death of Colonel Blimp* (1943)

Example: Color quantization

The image has 1280×936 pixels. Each pixel has red, green, and blue values = a point in $\{0, \dots, 255\}^3$. So we need

$$1280 \times 936 \text{ pixels} \times 3 \text{ bytes per pixel} = 3,594,240 \text{ bytes}$$

to store it.

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Idea: What if we instead cluster the pixels into $k = 16$ groups and replace each pixel by the nearest centroid (a representative color)?

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Idea: What if we instead cluster the pixels into $k = 16$ groups and replace each pixel by the nearest centroid (a representative color)?

With 16 groups, each pixel needs 0.5 bytes to name its group, and the 16 representative colors cost 48 bytes. Stored this way:

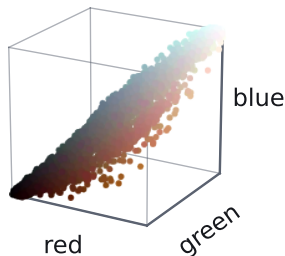
$$1280 \times 936 \text{ pixels} \times 0.5 \text{ bytes per pixel} + 48 \text{ bytes} = 599,088 \text{ bytes}$$

Example: Pixels as points

the photograph



its pixels, as points



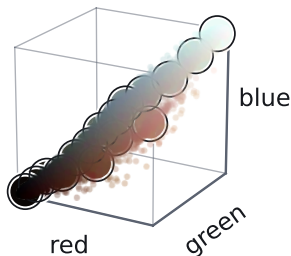
1,198,080 points in the “cube” $\{0, \dots, 255\}^3$, concentrated along its diagonal.

Example: Pixels as points

the photograph



its pixels, as points



Each pixel replaced by the nearest of 16 representative points.

Example: Image compressed by a factor of six

original, 108,909 colors

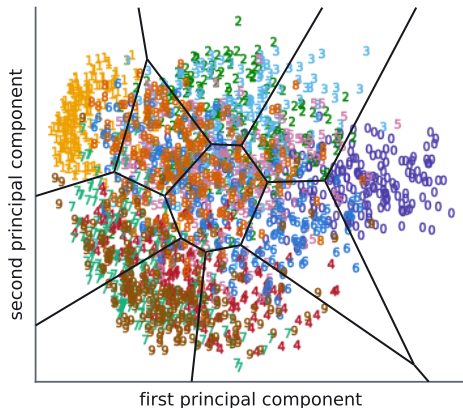


16 colors, chosen by k -means



Combining techniques: Do PCA, then cluster

The 2000 handwritten digits from hw1, each a point in $\mathbb{R}^{784}$.

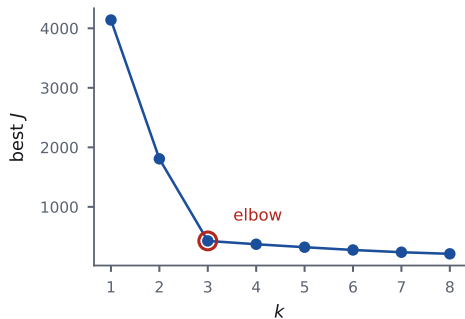


- keep the top **two** principal components capturing 17.6% of the variation
- run k -means++ with $k = 10$ on those two coordinates
- lines show the cluster boundaries

Plan for today

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- Probability review 15 min
- Mixtures of Gaussians 15 min

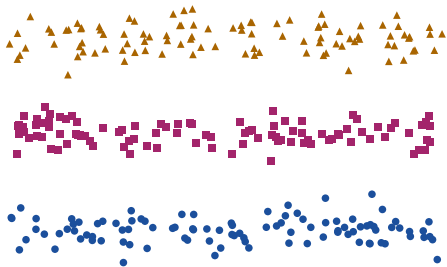
How do you choose k ?



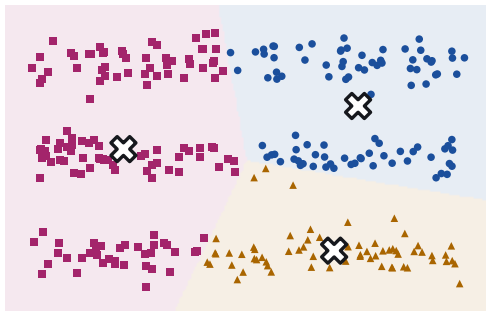
- J decreases every time k goes up, and reaches 0 at $k = n$
- So “choose k to minimize J ” always answers $k = n$
- Usual practice is to plot J against k and look for an **elbow**

Failure 1: Stripes with $k = 3$

the grouping we wanted

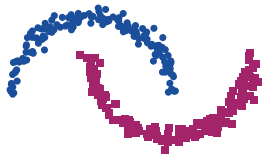


what k -means returns

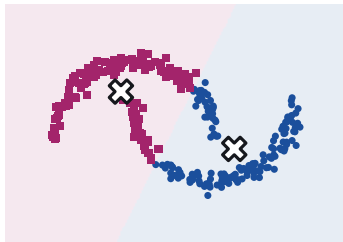


Failure 2: Non-convex clusters

the grouping we wanted



what k -means returns



Failure 3: No measure of confidence

The output is merely a label $c(i)$ for every point. What about:

- A point sitting halfway between two centers?
- A point far from every center?

Key idea: k -means has an objective, but we have no model of the process generating the data, hence we cannot provide a measure of confidence.

Plan for today

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Recall: Probability mass function

Definition (Probability mass function)

The probability mass function of a discrete random variable X is

$$p(x) = \mathbb{P}(X = x).$$

It satisfies $p(x) \geq 0$ and $\sum_x p(x) = 1$. We denote this by $X \sim p$.

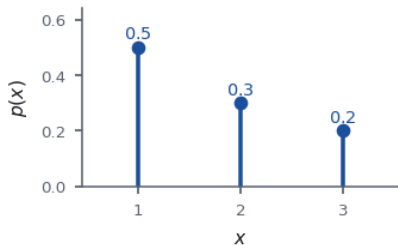
If A is an event, then

$$\mathbb{P}(X \in A) = \sum_{x \in A} p(x).$$

Example: Categorical distribution

If X takes values in $\{1, \dots, k\}$ according to $\mathbb{P}(X = j) = \pi_j$ for $\pi_j \geq 0$ and $\sum_{j=1}^k \pi_j = 1$, then we write

$$X \sim \text{Categorical}(\pi_1, \dots, \pi_k).$$



$$X \sim \text{Categorical}(0.5, 0.3, 0.2).$$

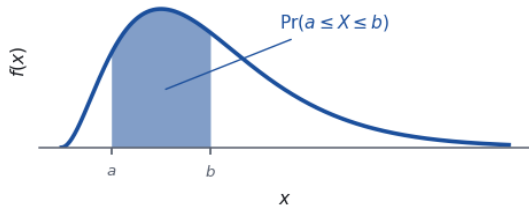
Definition: Probability density function

Definition (Probability density function)

The density of a real-valued random variable X is a function f satisfying

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1, \quad \text{and} \quad \mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx,$$

for all $a \leq b$. We denote this $X \sim f$.

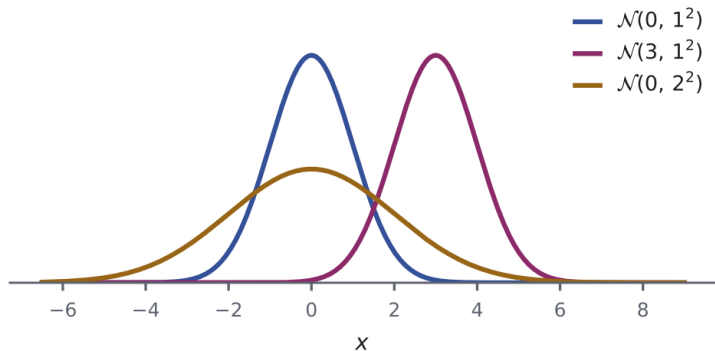


Watch out: The height of $f(x)$ is **not** a probability, area is.

Example: Gaussian distribution

A Gaussian random variable X with mean μ and variance σ^2 has density

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$



Recall: Three important rules

Let $p(x, y)$, $p(x | y)$, and $p(y | x)$ be the joint and conditional densities of two random variables, X and Y .

Definition (Sum, product and Bayes rules)

sum:
$$p(x) = \int_y p(x, y) dy$$

product:
$$p(x, y) = p(y | x)p(x)$$

Bayes:
$$p(x | y) = \frac{p(y | x)p(x)}{p(y)}$$

Watch out: ML literature often uses p to denote both p.m.f.s and p.d.f.s. Use context to determine its meaning.

Example: A mixture distribution

With probability 0.6, sample a standard Gaussian; with probability 0.4, sample a Gaussian with mean 5 and variance 1. Call the result X .

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One way to express X is through a **latent variable** Z :

$$Z = \begin{cases} 1 & w.p. \ 0.6 \\ 2 & w.p. \ 0.4 \end{cases}, \quad \text{i.e.,} \quad Z \sim \text{Categorical}(0.6, 0.4).$$

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We can then define X by specifying its conditional density given Z :

$$p(X \mid Z = 1) = \mathcal{N}(0, 1) \quad \text{and} \quad p(X \mid Z = 2) = \mathcal{N}(5, 1).$$

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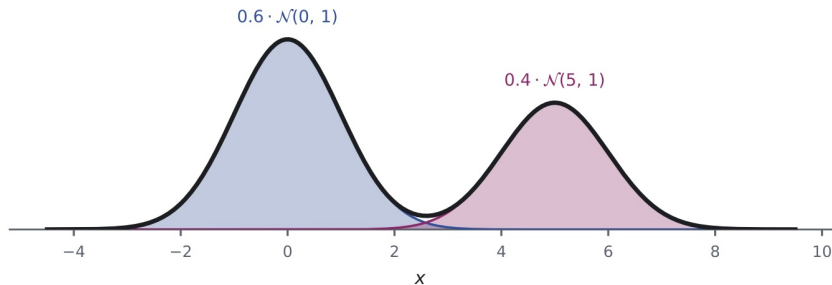
$$p(X \mid Z = 1) = \mathcal{N}(0, 1) \quad \text{and} \quad p(X \mid Z = 2) = \mathcal{N}(5, 1).$$

Question

What does the density of X look like?

Example: A mixture of distributions

With probability 0.6, sample a standard Gaussian; with probability 0.4, sample a Gaussian with mean 5 and variance 1. Call the result X .



Definition: A mixture distribution

Definition (Mixture)

Let $p_1, \dots, p_k$ be densities and let $\pi_1, \dots, \pi_k \geq 0$ with $\sum_j \pi_j = 1$. The **mixture** with components p_j and **weights** π_j is

$$p(x) = \sum_{j=1}^k \pi_j p_j(x).$$

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Let $p_1, \dots, p_k$ be densities and let $\pi_1, \dots, \pi_k \geq 0$ with $\sum_j \pi_j = 1$. The **mixture** with components p_j and **weights** π_j is

$$p(x) = \sum_{j=1}^k \pi_j p_j(x).$$

This is what you get from setting

$$p(x \mid Z = j) = p_j(x), \quad j = 1, \dots, k,$$

when $Z \sim \text{Categorical}(\pi_1, \dots, \pi_k)$.

Example: A mixture of k Gaussians

Given weights $\pi_1, \dots, \pi_k$ (non-negative and summing to 1), take every component to be a Gaussian:

$$p(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2).$$

Three numbers per component:

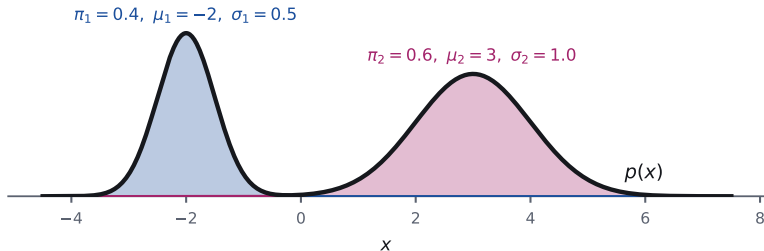
$\underbrace{\pi_j}$
how common?

$\underbrace{\mu_j}$
where?

$\underbrace{\sigma_j}$
how spread out?

Coming up: We can use mixtures of k (multivariate) Gaussians to model clusters of data points, and produce better clusterings than k -means!

Example: A mixture with $k = 2$



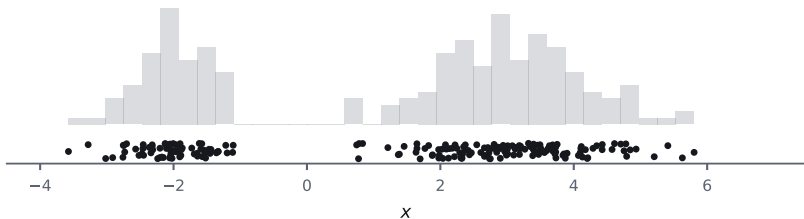
Key idea: The mixture density is obtained by **adding** the component densities, after weighting each one by how common it is.

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- **Mixtures of Gaussians** 15 min

Problem: Mixture modeling

Now suppose you are given $n = 200$ points.

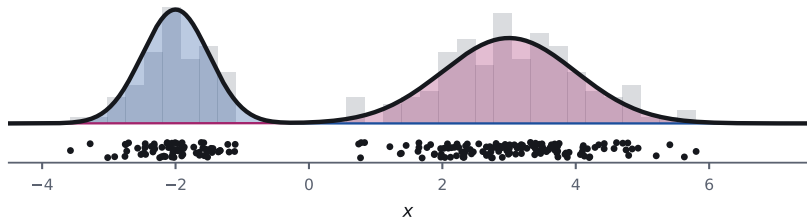


Question

What probability distribution might have generated these points?

Problem: Mixture modeling

Now suppose you are given $n = 200$ points.



One answer: a mixture of two Gaussians. But which two? The parameters are now **unknown**.

Definition: Gaussian mixture modeling

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$ and a number of components k , find weights $\pi_1, \dots, \pi_k$, means $\mu_1, \dots, \mu_k$, and variances $\sigma_1^2, \dots, \sigma_k^2$ such that

$$p(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2)$$

describes the data “well.”

Definition: Gaussian mixture modeling

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$$p(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2)$$

describes the data “well.”

Idea: To introduce what “well” will mean, let's look at the $k = 1$ case.

Special case: $k = 1$

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$, find a mean μ and variance σ^2 such that

$$p(x) = \mathcal{N}(x \mid \mu, \sigma^2)$$

describes the data “well.”

Special case: $k = 1$

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$, find a mean μ and variance σ^2 such that

$$p(x) = \mathcal{N}(x \mid \mu, \sigma^2)$$

describes the data “well.”

Key idea: We should choose μ and σ^2 to maximize the **likelihood** of observing the data we did.

Special case: $k = 1$

- If the data $x_1, \dots, x_n$ came from $\mathcal{N}(x \mid \mu, \sigma^2)$, then the **likelihood** of observing it would be

$$L(\mu, \sigma^2) = \prod_{i=1}^n \mathcal{N}(x_i \mid \mu, \sigma^2).$$

Special case: $k = 1$

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$$L(\mu, \sigma^2) = \prod_{i=1}^n \mathcal{N}(x_i \mid \mu, \sigma^2).$$

- Maximizers of L are maximizers of $\log L$ (**why?**), which equals

$$\sum_{i=1}^n \log \mathcal{N}(x \mid \mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Special case: $k = 1$

- If the data $x_1, \dots, x_n$ came from $\mathcal{N}(x \mid \mu, \sigma^2)$, then the **likelihood** of observing it would be

$$L(\mu, \sigma^2) = \prod_{i=1}^n \mathcal{N}(x_i \mid \mu, \sigma^2).$$

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- Taking derivatives with respect to μ and σ^2 shows that we should take

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

Back to Gaussian mixture modeling

Problem (Gaussian mixture modeling in 1D)

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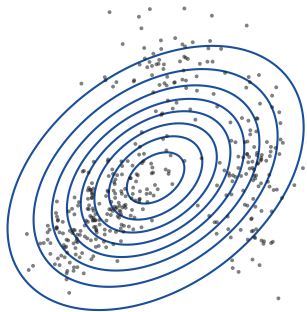
$$p(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2)$$

maximizes the likelihood.

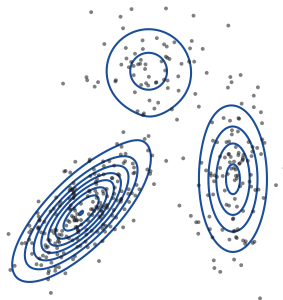
Coming up: We will generalize from $k = 1$ to general k , and then from univariate Gaussians to multivariate ones.

One Gaussian versus a mixture of three

one Gaussian



a mixture of three



Its highest density sits in the gap between the groups.

Definition: The multivariate Gaussian

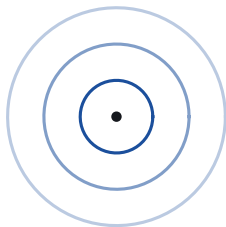
For $x, \mu \in \mathbb{R}^p$ and Σ symmetric and positive definite:

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right).$$

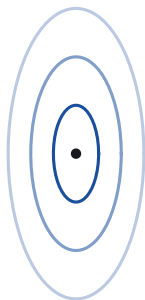
	in $\mathbb{R}$	in $\mathbb{R}^p$
the spread	σ^2 , one number	Σ , a $p \times p$ matrix
in the exponent	divide by σ^2	multiply by Σ^{-1}
out in front	$(\sigma^2)^{1/2}$	$ \Sigma ^{1/2}$

Three shapes of Σ

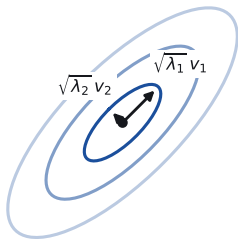
spherical $\Sigma = \sigma^2 I$



diagonal Σ



full Σ



spherical, $\Sigma = \sigma^2 I$
diagonal
full

1 number
 p numbers
 $p(p+1)/2$ numbers

round, one size
axis-aligned, one size per axis
any ellipse, in any orientation